

# S&T RAMS

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G-PRD S0600 REV.01

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| <b>GBMS</b>                | COUNTRY    |
|                            | <b>ALL</b> |
| DOCUMENT TYPE              | LANGUAGE   |
| <b>PROCESS DESCRIPTION</b> | <b>EN</b>  |

## REVISION HISTORY

| Rev. | Date     | Author          | Approver    | Authorizer | Revision Description                                                                                                                                                     |
|------|----------|-----------------|-------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00   | Mar 2022 | O. Samuel Espin | R. Castagna | P. Marmo   | First Issue                                                                                                                                                              |
| 01   | May 2023 | O. Samuel Espin | R. Castagna | P. Marmo   | Alignement with new GBMS Process Map. Updated interfaces with Service & Maintenance, S&T Asset Operation, S&T Manufacturing and S&T Requirements & Interface Management. |
|      |          |                 |             |            |                                                                                                                                                                          |
|      |          |                 |             |            |                                                                                                                                                                          |
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## DOCUMENTS TREE

## L1. G-PRD S0600 S&amp;T RAMS

L2. G-KPI S0601 S&amp;T RAMS KPIs

L2. G-PRC S0601 Management of Project RAMS and safety-related V&amp;V activities

L2. G-PRC S0602 Safety and Certification Activities

L3. G-PRC S0604 Safety-Related Application Conditions Management

L4. G-IND S0601 Guidelines for Safety-Related Application Conditions Management

L4. G-FOR S0601 Safety-Related Application Conditions Register

L2. G-PRC S0603 RAM &amp; LCC Activities Management

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## S&T RAMS

**G-PRD S0600**

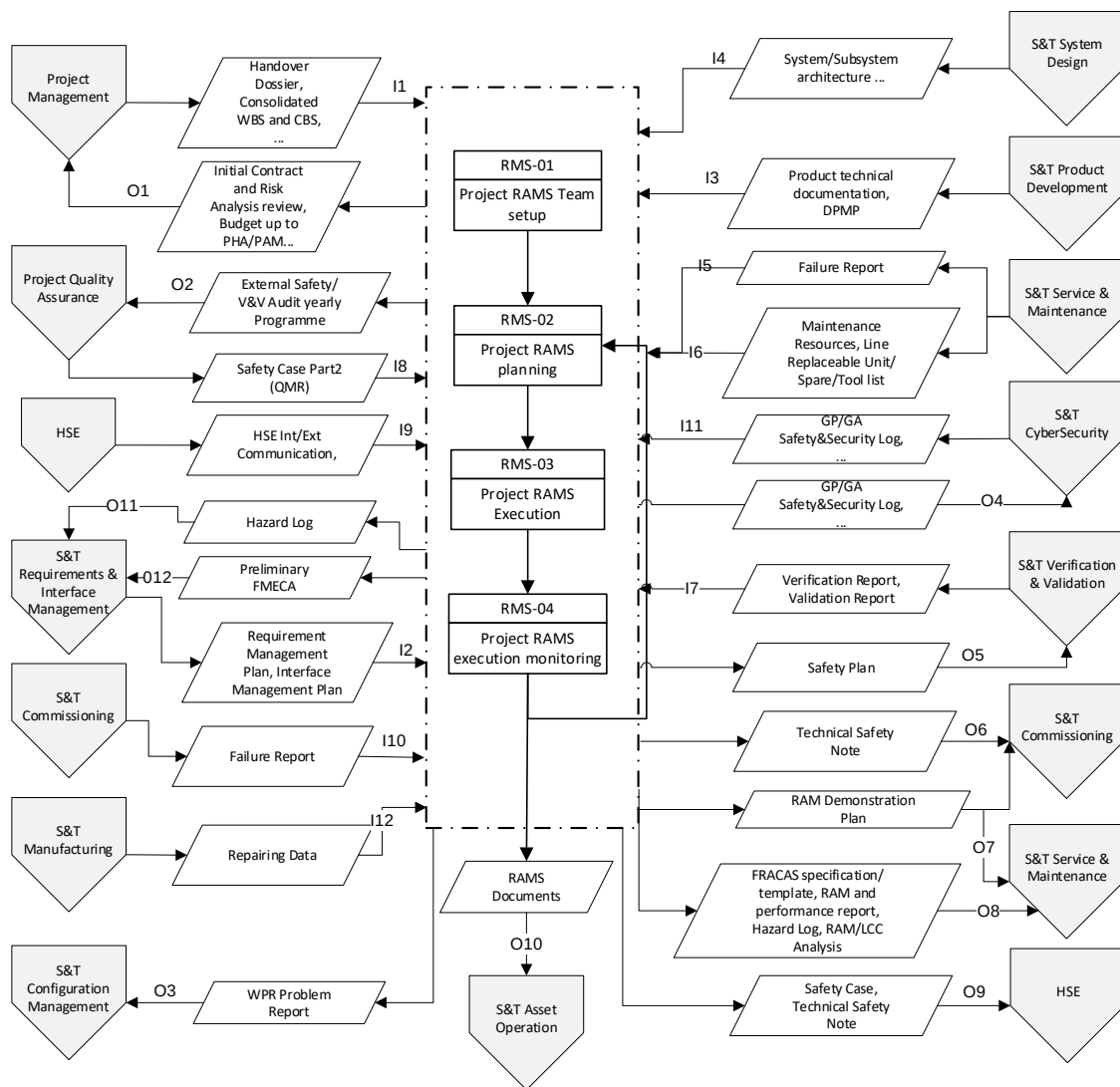
Rev. 01

**PROCESS CLUSTER:**

Execution

**PURPOSE**

The **RAMS** process shall demonstrate that a product or product component, a signalling system or subsystem, a turnkey system meets its specified Reliability, Availability, Maintainability, and Safety requirements, according to the applicable standards and contract constraints, and fulfils its intended use when placed in its intended environment. To ensure the correct implementation of Verification and Validation (V&V) governance, this process includes the monitoring of V&V activities.

**PROCESS DIAGRAM**


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| S&T RAMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | G-PRD S0600<br>Rev. 01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENTRY CRITERIA</b> <ul style="list-style-type: none"> <li>Availability of formalized releases of Contractual documents.</li> <li>Availability of formalized release of Project Management documentation.</li> <li>Availability of formalized release of technical documentation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <b>EXIT CRITERIA</b> <ul style="list-style-type: none"> <li>Fulfilment of all RAM and Safety requirements at System/ Subsystem Level.</li> <li>Gathering of all RAMS Reports.</li> <li>Fulfilment of RAM and Safety targets during revenue service (if applicable).</li> <li>Approval of contractual RAM and Safety Deliverables.</li> <li>Handover to the Customer/OS&amp;M.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>INPUTS</b> <ol style="list-style-type: none"> <li>During Project Launch/Planning: Handover Dossier, Consolidated WBS and CBS, Budget, Second Level Schedule, Project Management Plan.<br/>Iteratively during Project Execution: Updated WBS/CBS/budget, Risk/ Opportunity/Contingency, Second level schedule, Authorization of Change.</li> <li>Requirement Management Plan, Interface Management Plan.</li> <li>Product Technical Documentation (including HW/SW Architecture and Requirements), Development Project Management Plan (DPMP).</li> <li>System/Subsystem Architecture Definition and Requirements, Engineering Management Plan, ICDDs.</li> <li>Failure report.</li> <li>Maintenance Resources, Line Replaceable Unit/Spare/Tool list.</li> <li>Verification Report, Validation Report</li> <li>Safety Case Part2 (QMR)</li> <li>HSE Communication, HSE Reports.</li> <li>Failure Report</li> <li>GP/GA Safety &amp; Security Log, GP/GA Security Case, SA Safety &amp; Security Log, SA Security Case, Subsystem/System Safety and Security Log.</li> <li>Repairing Data</li> </ol> |  | <b>OUTPUTS</b> <ol style="list-style-type: none"> <li>During Project Launch/Planning: Initial Contract and Risk Analysis review, Budget up to PHA/PAM, Safety Plan, RAM Plan.<br/>Iteratively during Project Execution: Regular review of Project progress and deliverables, Functional Change Impact Estimation.</li> <li>External Safety/V&amp;V Audit yearly programme.</li> <li>WPR Problem Reports.</li> <li>GP/GA Safety &amp; Security Log Updated, SA Safety &amp; Security Log Updated, Updated Subsystem/System Safety and Security Log.</li> <li>Safety Plan.</li> <li>Technical Safety Note.</li> <li>RAM Demonstration Plan.</li> <li>FRACAS specification/template, RAM and performance report, [L] [SEP] Hazard Log, RAM/LCC Analysis</li> <li>Safety Case, Technical Safety Note.</li> <li>RAMS documents (including FRACAS procedures and Hazard Log).</li> <li>Hazard Log.</li> <li>Preliminary FMECA</li> </ol> |
| <b>ACTIVITIES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  | <b>REF.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>RMS-01: Project RAMS Team setup</b><br>The Project RAMS Manager (PRM) is appointed. The PRM attends project kick-off meeting organized by PM and gathers contractual documents and tender documents relevant to the project. Based on project Scope of Work, Work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | <b>G-PRC S0601</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| S&T RAMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              | G-PRD S0600<br>Rev. 01                    |
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| Packages and Project RAMS organization are defined. Project RAMS team starts working by analysing contract/project requirements.                                                                                                                                                                                                                                                                                                                                                                                                    |                                                              |                                           |
| <b>RMS-02: Project RAMS planning</b><br>The PRM coordinates with the RAM and Safety WPLs to detail the project schedule and the WBS, along with the risk analysis. RAMS WPLs estimate the budget and workload related to their own WP, keeping the PRM informed. In addition, they shall check if the V&V schedules and budgets are adequate to grant the proper execution of the V&V activities, V&V audit needs are planned. RAM and Safety WPLs define the RAM and Safety Plans.                                                 |                                                              | G-PRC S0601<br>G-PRC S0602<br>G-PRC S0603 |
| <b>RMS-03: Project RAMS execution</b><br>The PRM coordinates the execution of RAM and Safety activities, in order to achieve the expected results. The RAM and Safety WPLs are responsible to develop all RAM and Safety documents related to their owned WPs, based on project needs.                                                                                                                                                                                                                                              |                                                              | G-PRC S0601<br>G-PRC S0602<br>G-PRC S0603 |
| <b>RMS-04: Project RAMS execution monitoring</b><br>The PRM periodically monitors in dedicated meetings, the completeness and evolution of the project RAMS execution and related V&V activities, also involving the SIG WPLs, the System Validator and External Providers, if any. Updates for planning, monitoring, and control activities are detailed by RAM and Safety WPLs into the Project RAMS Report, which constitutes the official tool including all the main contents and information related to the project progress. |                                                              | G-PRC S0601                               |
| TITLES OF REF. DOCUMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                           |
| G-PRC S0601                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Management of Project RAMS and safety-related V&V activities |                                           |
| G-PRC S0602                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Safety and Certification Activities                          |                                           |
| G-PRC S0603                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RAM & LCC Activities Management                              |                                           |
| WORK ENVIRONMENT STANDARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                              |                                           |
| Teamcenter, IBM Doors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                              |                                           |
| TAILORING GUIDELINES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                              |                                           |
| None.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                              |                                           |
| MEASURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                              |                                           |
| Refer to <b>G-KPI S0601</b> that defines the KPIs associated with the process.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |                                           |
| VERIFICATION STEPS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                              |                                           |
| Process correct implementation is verified by means of audits performed by Quality Assurance or Internal Audit functions.                                                                                                                                                                                                                                                                                                                                                                                                           |                                                              |                                           |

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| S&T RAMS                   |                                                                                                                                                                                                                                         |                 |                                                          | G-PRD S0600  |
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|                            |                                                                                                                                                                                                                                         |                 |                                                          | Rev. 01      |
| R.A.C.I.                   |                                                                                                                                                                                                                                         |                 |                                                          |              |
| Action code                | R = Responsible                                                                                                                                                                                                                         | A = Accountable | C = Consulted                                            | I = Informed |
| RMS-01                     | PRM                                                                                                                                                                                                                                     | RAMS FMs        | PM, RAMS FMs                                             |              |
| RMS-02                     | RAMS WPLs, System Validator                                                                                                                                                                                                             | PRM             | PJC, PJP, WPLs SIG,                                      | PM           |
| RMS-03                     | RAMS WPLs                                                                                                                                                                                                                               | PRM             | System Validator, WPLs SIG, Maintenance Manager, S&M WPL | PM           |
| RMS-04                     | RAMS WPLs                                                                                                                                                                                                                               | PRM             | System Validator, WPLs SIG                               | PM           |
|                            |                                                                                                                                                                                                                                         |                 |                                                          |              |
| TERMS                      |                                                                                                                                                                                                                                         |                 |                                                          |              |
| Acronyms and Abbreviations |                                                                                                                                                                                                                                         |                 |                                                          |              |
| FM                         | Functional Manager                                                                                                                                                                                                                      |                 |                                                          |              |
| OS&M                       | Operation Service & Maintenance                                                                                                                                                                                                         |                 |                                                          |              |
| PJC                        | Project Controller                                                                                                                                                                                                                      |                 |                                                          |              |
| PJP                        | Project Planner                                                                                                                                                                                                                         |                 |                                                          |              |
| PM                         | Project Manager                                                                                                                                                                                                                         |                 |                                                          |              |
| PRM                        | Project RAMS Manager                                                                                                                                                                                                                    |                 |                                                          |              |
| V&V                        | Verification and Validation                                                                                                                                                                                                             |                 |                                                          |              |
| WP                         | Work Package                                                                                                                                                                                                                            |                 |                                                          |              |
| WPL                        | Work Package Leader                                                                                                                                                                                                                     |                 |                                                          |              |
| Definitions                |                                                                                                                                                                                                                                         |                 |                                                          |              |
| V&V Governance             | Set of practices, relations rules and tools, available to properly monitor the coherence of safety-related V&V activities, for a specific project, with the strategy derived from safety activities, the contract and applicable norms. |                 |                                                          |              |
| WPL SIG                    | WPL of Design and/or Generic Application and/or Specific Application and/or Verification and/or Validation related activities.                                                                                                          |                 |                                                          |              |

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